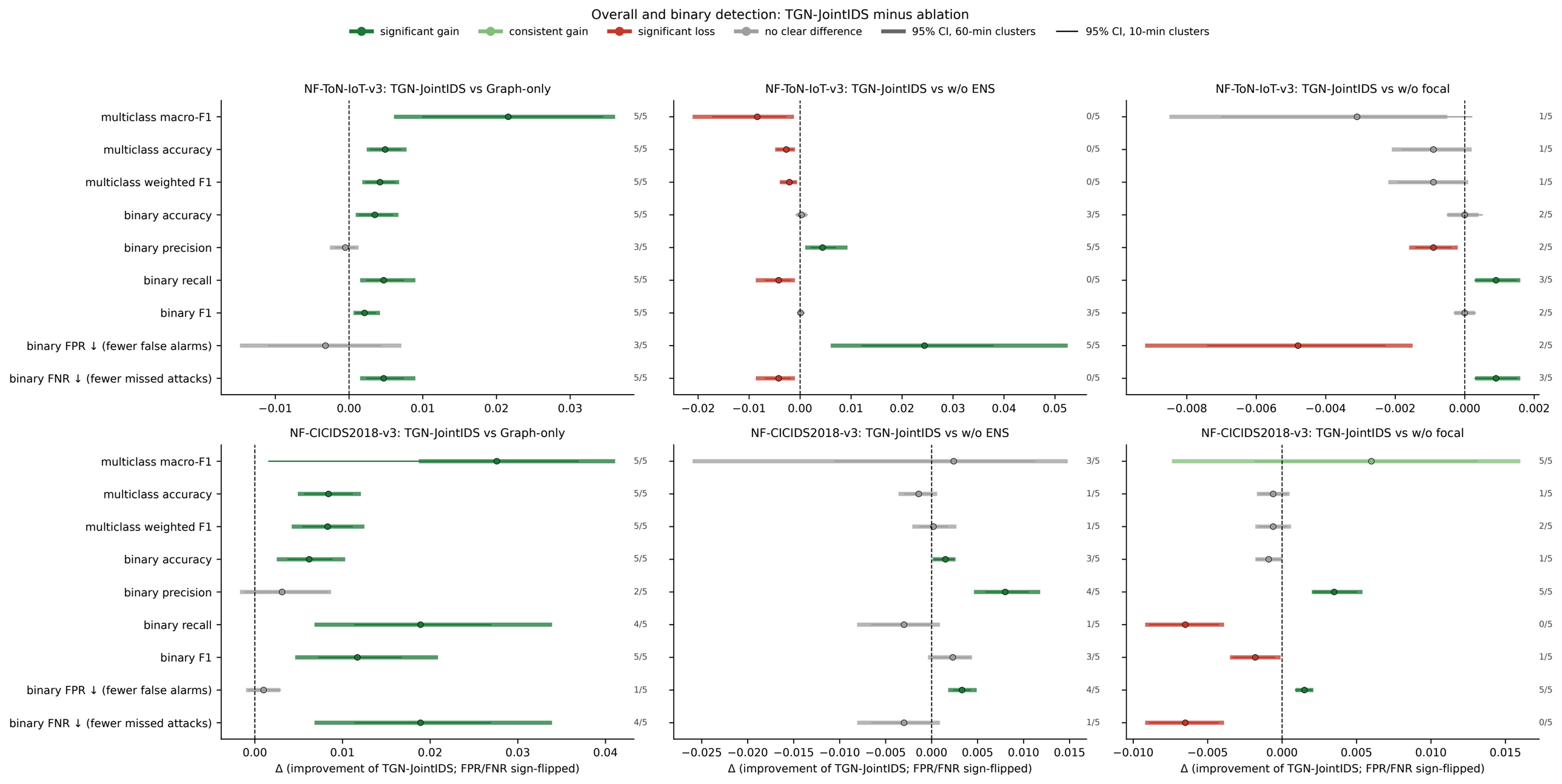
Overall and binary detection: TGN-JointIDS minus ablation
significant gain
consistent gain
significant loss
no clear difference
95% CI, 60-min clusters
95% CI, 10-min clusters
NF-ToN-IoT-v3: TGN-JointIDS vs Graph-only
NF-ToN-IoT-v3: TGN-JointIDS vs w/o ENS
NF-ToN-IoT-v3: TGN-JointIDS vs w/o focal
NF-CICIDS2018-v3: TGN-JointIDS vs Graph-only
NF-CICIDS2018-v3: TGN-JointIDS vs w/o ENS
NF-CICIDS2018-v3: TGN-JointIDS vs w/o focal
multiclass macro-F1
multiclass accuracy
multiclass weighted F1
binary accuracy
binary precision
binary recall
binary F1
binary FPR ↓ (fewer false alarms)
binary FNR ↓ (fewer missed attacks)
5/5
0/5
1/5
3/5
2/5
4/5
Δ (improvement of TGN-JointIDS; FPR/FNR sign-flipped)